

Supplementary Material: Adaptive Fidelity Selection in Multi-Objective Bayesian Optimisation for CFD-Based Engineering Design

Tinkle Chugh

Department of Computer Science, University of Exeter, Exeter, UK

1 Complete experimental protocol

The supplementary material documents settings and diagnostics that are useful for reproducibility but too detailed for the 15-page main paper. The final experiment comprises three design dimensions ($d = 2, 5, 10$), three optimisation methods (Adaptive MF, Fixed MF and HF-only), and ten independent seeds per combination, for 90 optimisation runs.

2 Numerical settings

All reported results use ten independent seeds per method and dimension. The three methods are Adaptive MF, Fixed MF and HF-only. The CFD model, geometry bounds and target-HF definition are identical across methods. Adaptive MF jointly maximises cost-normalised analytical EI over design and fidelity. Fixed MF uses the same modelling ingredients but requests fidelities according to the repeating $L, L, L, H, \dots$ schedule (nominally 25% HF), while HF-only requests HF at every sequential iteration. Initial designs are generated by feasible Latin hypercube sampling; the MF initial design contains a nested HF subset. The nominal EHF budgets are 12, 18 and 30 for $d = 2, 5, 10$, respectively. The sequential acquisition is optimised by CMA-ES with population/generation settings (10, 20), (14, 30) and (20, 45) for $d = 2, 5, 10$, respectively. GP hyperparameters are refitted for 100 Adam steps. The augmented Tchebycheff parameter is $\rho_{\text{aug}} = 0.05$, and 101 Das–Dennis weights are traversed in a seed-specific shuffled cycle.

The optimisation uses the qualification-derived normalised costs $(c_L, c_H) = (1, 14.5)$. Thus one HF evaluation costs 1 EHF and one LF evaluation costs $1/14.5 \simeq 0.069$ EHF. Uncached optimisation logs indicate that an HF CFD evaluation typically required about 9–10 min on the HPC nodes used for the study; the fixed cost ratio corresponds to an LF-equivalent cost of about 37–41 s. The acquisition uses the fixed qualification-derived ratio rather than noisy per-run wall-clock measurements. Quality escalations charge both the attempted LF solve and the subsequent HF solve, so realised expenditure can slightly exceed the nominal EHF budget. The accumulated expenditure is

$$C_{\text{EHF}} = \frac{n_L c_L + n_H c_H}{c_H},$$

and optimisation terminates when no admissible requested fidelity fits the remaining nominal EHF budget. Complete run-level fidelity counts, prediction errors, acquisition diagnostics, empirical-attainment plots and representative-design metadata are reported below.

Table 1. Core optimisation settings.

Setting	Value
GP covariance	ARD Matérn-5/2
GP mean	Constant
GP refit	100 Adam steps
Scalarisation	Augmented Tchebycheff
Augmentation	$\rho_{\text{aug}} = 0.05$
Weight set	101 Das–Dennis weights
Weight order	Seeded shuffle, cyclic traversal
Fixed-MF schedule	$L, L, L, H, \dots$ (nominal 25% HF)
EHF budgets ($d = 2, 5, 10$)	(12, 18, 30)
Initial MF LF/HF samples	(10/5), (15/5), (30/10) for $d = 2, 5, 10$
Initial HF-only samples	(5, 5, 10) for $d = 2, 5, 10$
Acquisition optimiser	CMA-ES
CMA-ES (λ, G_{max}), 2-D	(10, 20)
CMA-ES (λ, G_{max}), 5-D	(14, 30)
CMA-ES (λ, G_{max}), 10-D	(20, 45)
Geometry bounds	$3 \leq L/W_1 \leq 12$, $-0.08 \leq a_j \leq 0.08$
Maximum wall half-angle	25°

Quality control for low-fidelity In the current work, it is not sufficient to accept a requested LF result as a trustworthy observation, because a coarse numerical model can converge formally and still exhibit unacceptable variation in the quantity of interest or poor mass conservation. Therefore, a numerical-quality gate is applied after every LF solve. The postprocessor checks the stability of the pressure-loss coefficient over the final stored solver writes and the inlet-outlet flux balance. An LF result passes the objective-stability check when $\frac{\max K_p - \min K_p}{\max(|K_p|, \epsilon_K)} \leq 2 \times 10^{-3}$, where the maximum and minimum are calculated over the final stored values (when sufficient writes are available), and ϵ_K is a small numerical safeguard. The relative flux imbalance must additionally satisfy $\frac{|\phi_{\text{in}} + \phi_{\text{out}}|}{\max(|\phi_{\text{in}}|, |\phi_{\text{out}}|, \epsilon_\phi)} \leq 10^{-3}$. Here ϕ_{in} and ϕ_{out} denote the integrated inlet and outlet volumetric fluxes and ϵ_ϕ prevents division by zero.

If either quality condition fails, the same design $\mathbf{x}$ is evaluated at HF. This is called *LF-to-HF escalation*. Thus the requested and actual fidelities can be

written as

$$(m_t^{\text{req}}, m_t^{\text{act}}) = \begin{cases} (0, 0), & \text{LF requested and accepted,} \\ (0, 1), & \text{LF requested but escalated to HF,} \\ (1, 1), & \text{HF requested directly.} \end{cases}$$

An HF request is never downgraded to LF.

Geometric feasibility. The geometry validator rejects a design if any of the following holds:

1. non-positive local half-height;
2. local contraction when monotone expansion is required;
3. maximum local wall half-angle greater than 25° .

The production implementation uses this validator both during initial Latin-hypercube generation and inside CMA-ES. Infeasible acquisition candidates receive zero acquisition and are never sent to OpenFOAM.

3 CFD, mesh and fidelity settings

All fidelities solve the same steady incompressible RANS problem with `simpleFoam` and the standard k - ω SST model. The inlet width is $W_1 = 0.05$ m, area ratio is 2.5, inlet speed is 30 m/s, density is 1.225 kg/m^3 , kinematic viscosity is $1.5 \times 10^{-5} \text{ m}^2/\text{s}$, turbulence intensity is 5%, and turbulence length scale is $0.07W_1$. The Reynolds number based on inlet width is $Re_{W_1} = 10^5$.

Each proposed geometry is generated with `blockMesh`, checked using `checkMesh -allGeometry -allTopology`, and then solved using `simpleFoam`. The computational domain consists of two structured hexahedral blocks, one above and one below the centreline, with one cell in the spanwise direction. The front and back patches are `empty`; the upper and lower walls are no-slip. A uniform velocity and prescribed k - ω values are imposed at the inlet. Kinematic pressure is fixed to zero at the outlet and velocity and turbulence quantities use zero-gradient conditions there.

The production dictionaries use a cell-limited linear gradient, bounded linear-upwind convection for velocity, bounded upwind convection for k and ω , linear interpolation, and limited-corrected Laplacian and surface-normal-gradient schemes. Pressure is solved using GAMG, while velocity and turbulence variables use `smoothSolver`. One non-orthogonal corrector is used. Under-relaxation factors are 0.3 for pressure and 0.7 for U , k and ω .

M0 is the BO low fidelity, M2 is the target high fidelity, and M3 is used only as a finer numerical reference during qualification. LF residual-control targets are 10^{-5} for velocity, 10^{-4} for pressure and 10^{-5} for turbulence variables; HF targets are 10^{-6} , 10^{-5} and 10^{-6} , respectively. The fidelity hierarchy was assessed on six representative 2-D geometries spanning different lengths and wall perturbations. The optimisation subsequently uses M0 and M2 only; M3 is never queried by BO.

Table 2. Mesh hierarchy used to define and qualify fidelity.

Level	n_x	cells/half-height	wall grading	max SIMPLE iterations
M0	140	28	8	1500
M1	220	45	12	2500
M2	320	70	16	4000
M3	440	100	20	5000

Numerical-quality gate. A requested LF result is accepted only when the final stored pressure-loss values satisfy

$$\frac{\max K_p - \min K_p}{\max(|K_p|, \epsilon_K)} \leq 2 \times 10^{-3}$$

and the relative inlet-outlet flux imbalance satisfies

$$\frac{|\phi_{\text{in}} + \phi_{\text{out}}|}{\max(|\phi_{\text{in}}|, |\phi_{\text{out}}|, \epsilon_\phi)} \leq 10^{-3}.$$

If either condition fails, the same design is evaluated at HF. The failed LF objective is not treated as a trusted LF observation, but the attempted LF cost is retained. Requested and actual fidelities are therefore distinct:

$$(m_t^{\text{req}}, m_t^{\text{act}}) = \begin{cases} (0, 0), & \text{LF requested and accepted,} \\ (0, 1), & \text{LF requested but escalated to HF,} \\ (1, 1), & \text{HF requested directly.} \end{cases}$$

An HF request is never downgraded to LF.

4 Worked example of the analytical expected improvement

This example illustrates the calculation of the analytical EI used in the main paper. The numbers are illustrative and are not taken from a particular CFD run.

Suppose that, at a candidate design and a given fidelity, the GP prediction for the expensive objective is

$$\tilde{F}_1 \sim \mathcal{N}(0.4, 0.1333^2).$$

Consider the Tchebycheff weight

$$\mathbf{w} = (w_1, w_2) = (0.6, 0.4).$$

Then

$$A = w_1 \tilde{F}_1,$$

so

$$A \sim \mathcal{N}(\mu_A, \sigma_A^2),$$

with

$$\mu_A = 0.6(0.4) = 0.24, \quad \sigma_A = 0.6(0.1333) \approx 0.08.$$

Suppose the exact geometric objective is

$$f_2 = \frac{1}{3},$$

and therefore

$$b = w_2 \tilde{f}_2 = 0.4 \left(\frac{1}{3} \right) \approx 0.1333.$$

Using

$$\rho_{\text{aug}} = 0.05$$

and assuming the current scalar incumbent is

$$g^* = 0.33,$$

the scalarised random objective is

$$G = \begin{cases} 0.05A + 1.05b, & A \leq b, \\ 1.05A + 0.05b, & A > b. \end{cases}$$

Branch 1: $A \leq b$

For the first branch,

$$a_1 = 0.05, \quad c_1 = 1.05b \approx 0.14.$$

Positive improvement requires

$$A < \tau_1 = \frac{g^* - c_1}{a_1} = \frac{0.33 - 0.14}{0.05} = 3.8.$$

Since this branch is already restricted to

$$A \leq b = 0.1333,$$

the effective upper limit is

$$u'_1 = \min(0.1333, 3.8) = 0.1333.$$

The corresponding standard-normal coordinate is

$$z_1 = \frac{0.1333 - 0.24}{0.08} \approx -1.333.$$

Thus

$$P_1 = \Phi(-1.333) \approx 0.0918.$$

The truncated first moment is

$$M_1 = \mu_A \Phi(z_1) - \sigma_A \phi(z_1).$$

Using

$$\phi(-1.333) \approx 0.164,$$

we obtain approximately

$$M_1 = 0.24(0.0918) - 0.08(0.164) \approx 0.0089.$$

Hence

$$EI_1 = (g^* - c_1)P_1 - a_1 M_1$$

and therefore

$$EI_1 \approx (0.33 - 0.14)(0.0918) - 0.05(0.0089) \approx 0.017.$$

Branch 2: $A > b$

For the second branch,

$$a_2 = 1.05, \quad c_2 = 0.05b \approx 0.00667.$$

The improvement threshold is

$$\tau_2 = \frac{g^* - c_2}{a_2} = \frac{0.33 - 0.00667}{1.05} \approx 0.3079.$$

Thus improvement is possible over

$$0.1333 < A < 0.3079.$$

The corresponding standardised limits are

$$z_b = \frac{0.1333 - 0.24}{0.08} \approx -1.333,$$

and

$$z_{\tau_2} = \frac{0.3079 - 0.24}{0.08} \approx 0.849.$$

Hence

$$P_2 = \Phi(0.849) - \Phi(-1.333) \approx 0.711.$$

The truncated first moment is

$$M_2 = \mu_A P_2 + \sigma_A [\phi(z_b) - \phi(z_{\tau_2})].$$

Equivalently, direct substitution into the branch formula gives

$$EI_2 = [g^* - c_2 - a_2 \mu_A] P_2 + a_2 \sigma_A [\phi(z_{\tau_2}) - \phi(z_b)],$$

which is algebraically identical to $EI_2 = (g^* - c_2)P_2 - a_2M_2$. Using the standard-normal PDF values gives approximately

$$M_2 \approx 0.166.$$

Therefore

$$EI_2 = (g^* - c_2)P_2 - a_2M_2 \approx 0.060.$$

The total expected improvement is consequently

$$EI = EI_1 + EI_2 \approx 0.077.$$

If this candidate corresponds to LF with

$$c_L = 1,$$

then

$$\alpha_L = \frac{EI_L}{c_L} \approx 0.077.$$

Suppose the same design evaluated at HF has

$$EI_H = 0.40$$

but costs

$$c_H = 14.5.$$

Then

$$\alpha_H = \frac{0.40}{14.5} \approx 0.0276.$$

Although HF has the larger *raw* expected improvement,

$$EI_H > EI_L,$$

the cost-normalised values satisfy

$$\alpha_L > \alpha_H.$$

The acquisition would therefore request LF at this design.

This example illustrates the central fidelity-selection mechanism: the method does not simply choose the fidelity with the largest predicted improvement, but the fidelity offering the largest expected scalarised improvement per unit qualified computational cost.

5 Run-level final results

The p -values are from two-sided Wilcoxon signed-rank tests on paired seeds. Only the 2-D Adaptive–Fixed comparison is significant at the 5% level.

Table 3. Median [IQR] final hypervolume and median realised EHF from ten runs.

d	Method	Final HV	EHF
2	Adaptive MF	18.669 [17.668,18.954]	12.41
2	Fixed MF	18.101 [17.212,18.820]	12.38
2	HF-only	18.997 [18.116,19.011]	12.00
5	Adaptive MF	19.067 [18.916,19.186]	18.21
5	Fixed MF	18.860 [18.730,19.084]	17.76
5	HF-only	19.189 [19.087,19.229]	18.00
10	Adaptive MF	19.215 [19.172,19.250]	30.41
10	Fixed MF	19.253 [19.236,19.283]	30.21
10	HF-only	19.296 [19.228,19.327]	30.00

Table 4. Final repeated-seed results. Hypervolume is median [IQR]. EHF is the median realised equivalent-HF expenditure. The last three columns are median run-level fractions during the sequential phase: HF requested by the policy, HF actually evaluated after quality control, and LF requests escalated to HF.

d	Method	Final HV	EHF	Req. HF	Actual HF	Escalation
2	Adaptive MF	18.669 [17.668,18.954]	12.41	0.26	0.40	0.13
2	Fixed MF	18.101 [17.212,18.820]	12.38	0.21	0.45	0.23
2	HF-only	18.997 [18.116,19.011]	12.00	1.00	1.00	0.00
5	Adaptive MF	19.067 [18.916,19.186]	18.21	0.28	0.49	0.17
5	Fixed MF	18.860 [18.730,19.084]	17.76	0.22	0.39	0.17
5	HF-only	19.189 [19.087,19.229]	18.00	1.00	1.00	0.00
10	Adaptive MF	19.215 [19.172,19.249]	30.41	0.39	0.50	0.11
10	Fixed MF	19.253 [19.236,19.283]	30.21	0.23	0.41	0.18
10	HF-only	19.296 [19.228,19.327]	30.00	1.00	1.00	0.00

6 Fidelity allocation diagnostics

The requested-HF fraction records the algorithmic fidelity decision, whereas the actual-HF fraction records the fidelity ultimately executed after the numerical-quality gate. This distinction is particularly important for Fixed MF: although its requested schedule is nominally 25% HF, escalation can raise its realised HF fraction substantially.

7 Surrogate and acquisition diagnostics

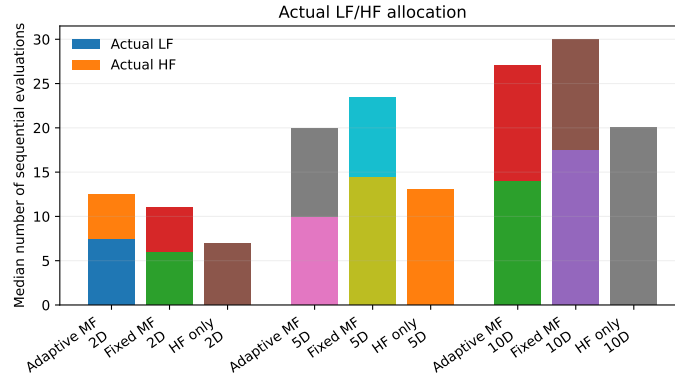
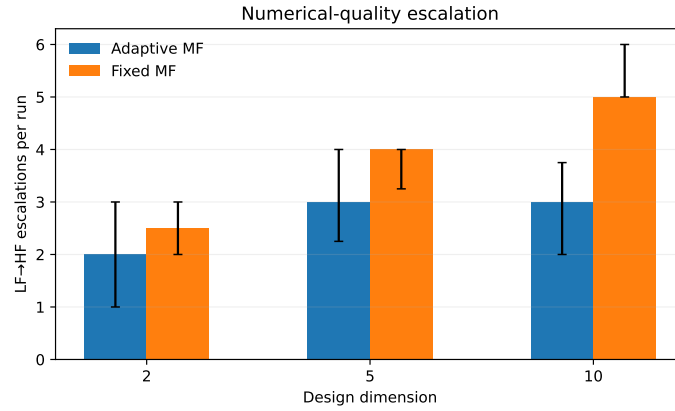
See Figure 4 and 5.

8 Empirical attainment plots

See Figures 6, 7

Table 5. Paired seed-wise hypervolume comparisons involving Adaptive MF.

d	Comparison	Median difference	Wins/10	p
2	Adaptive–Fixed	+0.095	7	0.0488
5	Adaptive–Fixed	+0.093	7	0.3223
10	Adaptive–Fixed	−0.026	2	0.1602
2	Adaptive–HF-only	−0.139	3	0.1602
5	Adaptive–HF-only	−0.039	2	0.1934
10	Adaptive–HF-only	−0.047	2	0.2324

**Fig. 1.** Median LF/HF evaluation counts by method and dimension.**Fig. 2.** Summary of LF-to-HF numerical-quality escalations.

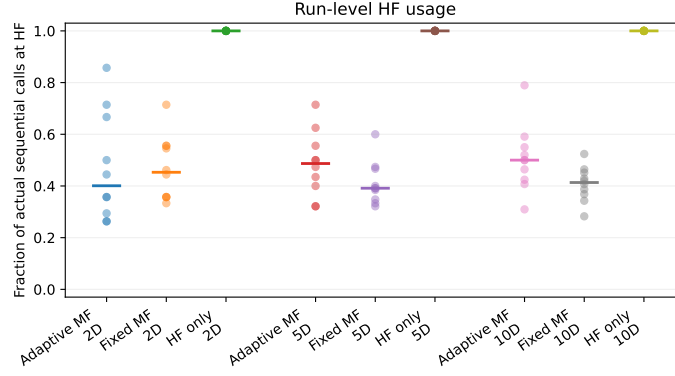


Fig. 3. Run-level fraction of sequential evaluations realised at HF.

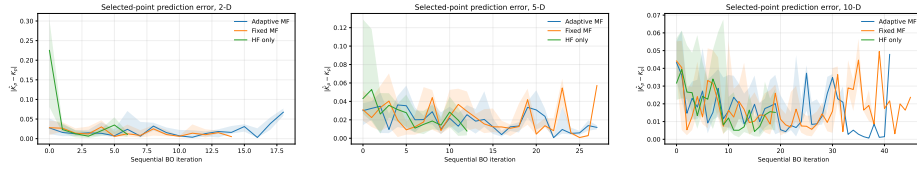


Fig. 4. Absolute pressure-loss prediction error over BO iterations.

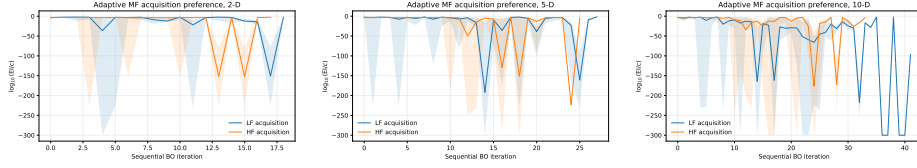


Fig. 5. Evolution of the cost-normalised LF and HF acquisition values for Adaptive MF.

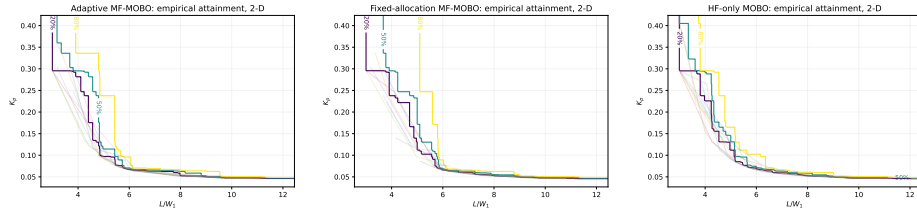


Fig. 6. 2-D empirical attainment plots: Adaptive MF, Fixed MF and HF-only.

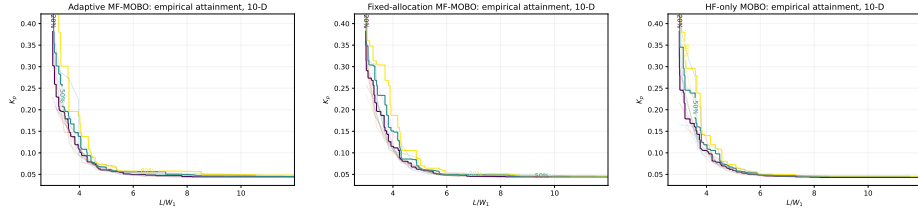


Fig. 7. 10-D empirical attainment plots: Adaptive MF, Fixed MF and HF-only.